

LESSON**1****How an FTC TeleOp Program Works***Become the program and trace the complete path***MISSION** Can you move the program token from INIT to STOP without skipping a step?**Learning targets**

CHECK	BY THE END OF THIS LESSON, I CAN...
[]	Name the program's four main phases.
[]	Trace the path from hardware mapping to the active control loop.
[]	Explain why the loop repeats until STOP.
[]	State the six-step POWERING ON sequence.

Words to know**VOCABULARY** OpMode | LinearOpMode | hardwareMap | INIT | PLAY | control loop | telemetry | STOP**Code focus**

```

leftMotor = hardwareMap.get(DcMotor.class, "left_motor");
waitForStart();
while (opModeIsActive()) {
    // READ -> COMMAND -> REPORT -> REPEAT
}

```

Before the video: commit to a prediction

MY PREDICTION Circle the phase where you think the program spends most of its driving time: INIT / WAIT / ACTIVE LOOP / STOP

Confidence before the lesson: 1 2 3 4 5 (circle one)

The POWERING ON safety contract

NON-NEGOTIABLE The sequence is completed before every live-motor test. If a step is skipped, remain disabled and restart at Step 1.

STEP	ACTION	TEAM CHECK
1	SECURE THE ROBOT Use stable blocks. Keep driven wheels and moving mechanisms clear of the table and floor.	[]
2	CONFIRM DISABLED Select the correct robot and OpMode, but do not press PLAY.	[]
3	LOOK AROUND Remove hands, tools, wires, scraps, and loose objects from moving areas.	[]
4	MAKE SURE ALL ARE CLEAR Everyone steps outside the boundary; secure hair, clothing, lanyards, strings, and jewelry.	[]
5	CALL AND WAIT Call POWERING ON! loudly. Pause long enough for everyone to stop, look up, and step away.	[]
6	VISUALLY CONFIRM, THEN ENABLE Scan the full area again. Press PLAY only when everyone and everything is clear.	[]

Callout script

BEFORE PLAY Robot is secured and DISABLED. Look around. Everyone clear? POWERING ON! ... Wait ... Area is visibly clear. Enabling now.

BEFORE CONTACT STOP pressed. Driver Station shows DISABLED. Hands may approach only after the operator confirms disabled.

Student agreement

I understand that anyone may call STOP and the operator will disable immediately. I will not approach or touch the robot until DISABLED is visually confirmed.

Student signature: _____ Date: _____

Reference program: BasicTwoMotorTeleOp.java

Use this numbered reference throughout Lessons 1-6. The comments divide the program into startup and repeating control sections.

```

01 package org.firstinspires.ftc.teamcode;
02
03 import com.qualcomm.robotcore.eventloop.opmode.LinearOpMode;
04 import com.qualcomm.robotcore.eventloop.opmode.TeleOp;
05 import com.qualcomm.robotcore.hardware.DcMotor;
06
07 @TeleOp(name="Basic Two Motor TeleOp", group="Iterative Opmode")
08 public class BasicTwoMotorTeleOp extends LinearOpMode {
09     private DcMotor leftMotor;
10     private DcMotor rightMotor;
11
12     @Override
13     public void runOpMode() {
14         leftMotor = hardwareMap.get(DcMotor.class, "left_motor");
15         rightMotor = hardwareMap.get(DcMotor.class, "right_motor");
16
17         // Set one motor REVERSE if the physical drivetrain requires it.
18         // rightMotor.setDirection(DcMotor.Direction.REVERSE);
19
20         telemetry.addData("Status", "Initialized");
21         telemetry.update();
22         waitForStart();
23
24         if (opModeIsActive()) {
25             while (opModeIsActive()) {
26                 double leftPower = -gamepad1.left_stick_y;
27                 double rightPower = -gamepad1.right_stick_y;
28                 leftMotor.setPower(leftPower);
29                 rightMotor.setPower(rightPower);
30                 telemetry.addData("Left Power", leftPower);
31                 telemetry.addData("Right Power", rightPower);
32                 telemetry.update();
33             }
34         }
35     }
36 }

```

CODE DETECTIVE Circle `waitForStart()`. Box the while condition. Underline both `setPower()` calls. Draw a star beside each telemetry line.

LESSON

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Guided Video Notes

How an FTC TeleOp Program Works

DIRECTIONS At each timestamp, pause, think silently, discuss, and then record evidence from the code or whiteboard visual.

PAUSE	QUESTION	MY CODE / VISUAL EVIDENCE
0:18	What are the program's four main stations?	
0:47	Do declared motor variables already control physical motors?	
0:58	Why must configuration names match quoted strings exactly?	
1:06	What does waitForStart() do?	
1:26	What must happen before enabling?	
1:37	What four actions repeat inside the loop?	
1:48	Why does the joystick code contain a minus sign?	

MOST IMPORTANT IDEA The idea I would teach to a teammate is:

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1

Hands-On Team Activity

How an FTC TeleOp Program Works

Materials and setup

READY	ITEM
[]	Phase cards: IDENTIFY, CONNECT, WAIT, READ, SET POWER, TELEMETRY, REPEAT, STOP
[]	Program token and INIT / PLAY / STOP cards
[]	Joystick-value cards and a printed copy of BasicTwoMotorTeleOp.java
[]	POWERING ON safety poster (live robot is optional for this activity)

Team roles

ROLE	JOB
Program token	Shows the instruction currently being executed.
Driver Station	Announces INIT, PLAY, and STOP.
Hardware map	Connects Java names to configured devices.
Wait gate	Holds the token until PLAY.
Joystick reader	Supplies a fresh value every cycle.
Motor command	Receives and sends the requested powers.
Telemetry	Reports current commanded values.
Active check	Chooses REPEAT when true and STOP when false.

Activity sequence

STEP	ACTION
1	Arrange IDENTIFY -> CONNECT -> WAIT. Keep the token at WAIT until the Driver Station says PLAY.
2	After PLAY, pass the token through READ -> SET POWER -> TELEMETRY -> REPEAT.
3	Return to the active check. Show TRUE and begin another cycle with a new joystick card.
4	Complete three cycles. Explain why the newest input is used each time.
5	Show STOP/FALSE. The token must exit the loop and move to DISABLED.
6	Point to the safety poster and rehearse the six steps before any future live test.

LESSON**1****Lab Record and Exit Ticket***How an FTC TeleOp Program Works*

Student/team: _____ Date: _____ Robot: _____

Human TeleOp flow record

TOKEN LOCATION	WHAT HAPPENS	ONCE OR REPEATED?	MY EVIDENCE
IDENTIFY	Find the selected OpMode	Once	
CONNECT	Map Java names to hardware	Once	
WAIT	Pause for PLAY	Once	
READ	Get current joystick values	Repeated	
SET POWER	Send commands to motors	Repeated	
TELEMETRY	Report values	Repeated	
STOP	Exit loop and disable	At stop	

Debrief

#	QUESTION	MY RESPONSE
1	Which actions happen only once before PLAY?	
2	Which actions repeat during TeleOp?	
3	Why does the program return to READ instead of reusing an old value?	
4	What condition sends the token to STOP?	

Exit ticket

ANSWER BEFORE LEAVING Write the complete TeleOp path in one sentence, beginning with INIT and ending with STOP.
